

Lab #:

Date:

Title: Thermometric Titration

Topic: Volumetric Analysis

Aim: To determine the concentration of a solution of hydrochloric acid by titrating it against a standard solution of 1 M sodium hydroxide using thermometric titration.

Apparatus/Materials:

Diagram:

Method:

1. Fill the burette with the sodium hydroxide solution.
2. Place a plastic cup with lid on a heat-resistant surface.
3. Attach a thermometer to the cup via a hole in the lid.
4. Use a measuring cylinder to transfer 40 mL of hydrochloric acid solution to the plastic cup.
5. Use the thermometer to measure the initial temperature of the hydrochloric acid solution.
6. Record the initial temperature of the hydrochloric acid solution in the plastic cup.
7. Slowly add 5 mL of the sodium hydroxide solution from the burette to the hydrochloric acid solution in the plastic cup.
8. Stir the solution gently and measure the temperature of the reacting mixtures.
9. Repeat the process of adding 5 mL of the sodium hydroxide solution to the hydrochloric acid solution and measuring the temperature until a total volume of 50 mL is added.
10. Plot a graph of temperature of the solution against the volume of the sodium hydroxide solution added.

Observations:

Results:

Trial	Volume of NaOH added/mL	Temperature of solution/°C
1	5.00	
2	10.00	
3	15.00	
4	20.00	
5	25.00	
6	30.00	
7	35.00	
8	40.00	
9	45.00	
10	50.00	

Calculations:

Discussion:

Sources of Error:

Limitations:

Precautions:

Conclusion:

Treatment of Results:

1. Plot a graph of Temperature (vertical axis) against volume of sodium hydroxide added (horizontal axis).
2. Draw two straight lines of best fit and extend them until they cross. The point at which the two lines meet corresponds to the volume of the sodium hydroxide needed for neutralization and to the maximum temperature.
3. Identify and report the Equivalence volume from the graph.
4. Identify and report the maximum temperature reached from the graph.
5. Use the information from your graph to calculate the concentration of the hydrochloric acid solution.
6. Describe and explain the trend in temperature change throughout experiment.
7. Explain the relationship between the shape of the graph and the process of neutralization.
8. Include sources of error and explain the effect that they have on the results.
9. Include limitations and explain the effects that they have on the results.
10. Include precautions and ensure that they are correctly justified in relation to procedure, accuracy, reliability or safety.
11. Write a suitable conclusion.